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RESEARCH

Automated Classification of Biscuit Quality Using YOLOv8 Models in Food Industry

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Abstract
It is of great importance for food safety and consumer satisfaction that industrial food products are durable, hygienic, and flawless. Robust products protect the physical integrity of the product by preventing damage that may occur during the production and transportation processes, which meets the expectations of the consumer. Hygienic production conditions prevent foodborne diseases by minimizing the risk of microbial contamination and protect consumer health. Perfect products strengthen the brand image with their aesthetic and satisfactory features and increase consumer loyalty. In the study conducted in this context, the classification of defect and no defect conditions of biscuits in the food industry was examined using YOLOv8 models. A summary dataset consisting of 4990 biscuit images was created and the biscuits were initially divided into two categories: defect and no defect. Later, defect biscuits were classified into three subcategories: not complete, overcooked, and texture defect. As a result of experiments with YOLOv8 models, binary classification (defect, no defect), the highest accuracy rate was achieved in the YOLOv8-m, YOLOv8-l, and YOLOv8-x models with 96.78%, while the highest accuracy rate in the triple classification (not complete, overcooked, and texture defect) performance was achieved in the YOLOv8-m model with 96.99%.

Keywords Biscuit classification · Deep learning · YOLOv8

Introduction

In the industry, the protection of the quality and safety of food products is of great importance. Food quality and safety have become an increasingly important issue today. As the demands of consumers increase, the application of effective methods for detecting and separating unwanted elements becomes inevitable (Taspinar 2022).

During production, products may be damaged, foreign objects may be mixed into the products on the production line, production may be disrupted, and sometimes products may be exposed to excessive heat. Such foreign substances, defective products pose serious risks in terms of food safety. Food products can have acrylamide if carbohydrate-rich foods are cooked at high temperatures for too long (Özkaynak & Ova 2006). It is known that acrylamide can cause cancer in humans, so it should not be eaten (Özkaynak & Ova 2006). Effective monitoring and control of production processes is vital to ensuring the safety and quality of food products. Food industry inspections help identify problematic products and foreign materials in a timely manner, especially in this context.

This study aims to develop a classification model capable of detecting biscuit damage effectively. This method can be used to identify and classify damaged biscuits automatically. In order to solve the biscuit defects, the image processing method is used. Also, the results from the study will create a dataset that can be used for data mining and artificial intelligence methods. Using a total of 4990 images, the biscuits were divided into two separate categories in the first place: defect biscuit, no defect biscuit. Then, the defect biscuits are divided into three groups so that there will be overcooked, not complete, and texture defect error in them. Classification operations were performed with YOLOv8-and, YOLOv8-s, YOLOv8-m, YOLOv8-l, and YOLOv8-x models on this

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dataset (Jiang et al. 2024; Liu et al. 2023). In this study, the obtained performance metrics, confusion matrix results, and related graphs were examined comparatively. In the model training processes, the accuracy and train/loss graphs of each YOLOv8 model were accessed. The confusion matrix analysis visually presented the correct and incorrect classification rates of each biscuit type and thus determined which types of models were more successful or inadequate. In the following sections of the study, the methods used, the findings obtained, and the discussion sections, respectively, are discussed in detail. The goal is to contribute to the production of high-quality and robust products.

Related Works

Srivastava et al. (2014) used six thresholding methods such as ISO data, minimum error, memory preservation, fuzzy logic, manual method, and k-mean clustering to extract chocolate chips from cookie images taken in their study. Image processing operations such as entropy, area, parameter calculation, and rigidity have been applied on Krack-Jack commercial biscuits. The proposed algorithm is able to detect cracks and stains and is equipped with a low-cost machine vision system. These algorithms, developed with MATLAB 5.2, have been tested on commercial biscuits and have achieved a specificity of more than 94% and a sensitivity of more than 82%.

Nashat and Abdullah (2010) proposed an automatic and intelligent system for the color control of biscuit products in their study. Advanced classification techniques such as support vector machines (SVM) and Wilk's λ analysis were used in this system. As a result of the experiments conducted, it was found that the system reached an accuracy rate of 87.25% for the directed acyclic graph (DAG) classifier and 86.75% for the binary boosted tree (BBT) classifier. In addition, it has been stated that the software-based application of the system is also applicable to other bakery products.

Wang et al. (2015) have developed a system for detecting biscuit defects using computer vision technology. In his studies, the focus was on the detection of defects such as partial deletion and overflow of cream. For this purpose, a system consisting of two cameras and innovative algorithms were used. As a result of online experiments, it has been determined that the system performs defect detection with high accuracy with low error and false alarm rates.

Khan et al. (2017) proposes a biologically inspired system for automatic and intelligent inspection of bakery products. This system includes advanced classification techniques such as HMAX-based shape identifier and support vector machine. With this method, it has been determined that baking products can be classified according to different cooking times based on their shapes and colors. It was determined that the one-versus-one support vector machine (SVM) and the directed cyclic graph support vector machine (SVM) achieved the highest accurate classification rate. The proposed method has achieved 95% and 100% classification accuracy for biscuit shape recognition and biscuit color recognition, respectively.

Paquet-Durand et al. (2012) demonstrates the development of an identification system using the modified Viola Jones algorithm for bread pieces. This system has achieved 94.4% accuracy with the developed version. In addition, a number of color properties and surface texture properties have been developed to test sensitivity to appearance changes and material composition variations in baked goods such as bread and biscuits.

In Yeh et al. (1995), RGB color space and a feed-forward propagation artificial neural network were used to monitor the color changes of biscuit samples during the baking process. The results show that the proposed algorithm provides 40% better recognition accuracy compared to manual visual inspection.

Trafialek et al. (2016) investigated the causes of foreign objects mixed into products on the production line. They found that foreign objects were mixed into the products during the transportation process of raw materials and packaging of products.

Nashat et al. (2014) proposed a machine learning–based method for automatic detection of cracks in biscuits in their study. They have extracted the characteristics of cracked and non-cracked biscuit images. They classified using the SVM algorithm. With the method they suggested, they detected cracks in biscuits with 97% accuracy.

Nashat and Abdullah (2016) examined irregularities, damages, and discolorations in bakery products of various colors, shapes, and sizes. They suggested that product defects can be detected faster with artificial intelligence systems without the need for human power.

Núria Banús et al. (2021a, b) have suggested the use of computer vision and artificial intelligence techniques by focusing on the importance of quality control in the packaging process in the food industry. In this context, they especially focused on the effectiveness of convolutional neural network (CNN)–based methods and the selection of deep learning models. These studies emphasize the importance of selecting the appropriate deep learning model and using pre-trained networks, especially for an important industrial need. In this context, in the proposed study, they have shown how computer vision technology can be used effectively to automate quality control in production processes.

Nuria Banús et al. (2021a, b) proposes a system for automating packaging control in industrial pizza production using computer image and deep learning techniques. They have determined that a specially developed ResNet18 CNN performs better than commercial off-the-shelf solutions. The system has increased the production speed by supporting pizzas of different brands and materials and has adapted to different densities and silk-screen patterns. In the tests performed, it was seen that false positive rates varied between 0.02% and 0.25%, and false negative rates varied between 0% and 0.08%.

Kim et al. (2021) used supervised learning to detect problems in X-ray images of used foods in real time. They created a training dataset by adding foreign objects to normal product images. Using this data and the YOLOv4 network, they detected foreign particles in the test data. This method was tested on images of pasta, snacks, pistachios, and red beans. The results showed that normal and faulty products were classified with at least 94% accuracy for all packaged foods. Their method also performed well in detecting foreign objects, which are usually hard to find with traditional methods and unsupervised learning.

In Taspinar's (2022) conference paper, he focused on image processing techniques for the classification of biscuits. The results obtained using the VGG16 deep learning architecture have reached an important level of accuracy in the classification of biscuit images. With the application of the methods performed, artificial neural networks and logistic regression were used to achieve accuracy rates of 98% and 97.4%, respectively.

In this work, an image dataset was created for the classification of biscuits. Using a total of 4990 images, biscuits were first classified into two categories: defect and no defect. The defect biscuits are divided into three groups as overcooked, not complete, and texture defect. Classification was made with YOLOv8-n, YOLOv8-s, YOLOv8-m, YOLOv8-l, and YOLOv8-x models. Performance metrics, confusion matrix results, and related graphs have been compared. The accuracy and train/loss graphs of each model have been achieved. Performance metrics, confusion matrix results, and related graphs have been compared. The accuracy and train/loss graphs of each model have been achieved. In this way, it was determined which types of models were more successful or inadequate. The methods used in the study, the findings obtained, and the discussion were thoroughly examined. The goal is to contribute to the production of high-quality and reliable products.

Material and Methods

This study aims to classify the biscuit, which is a part of the food industry, by investigating the defect and no defect. An image dataset has been created for the classification of biscuits. A classification process was carried out using YOLOv8 models with the data obtained from 4990 biscuit images. Biscuits were divided into two separate categories in the first place: defect biscuit, no defect biscuit. Then, the defect biscuits are divided into three groups so that there will be overcooked, not complete, and texture defect error in them. In Fig. 1, it shows a graphical representation of the proposed study. Thanks to this innovative approach, the study makes it easier to obtain better quality and robust products as a result of integrating YOLOv8 models with their applications in the food industry.

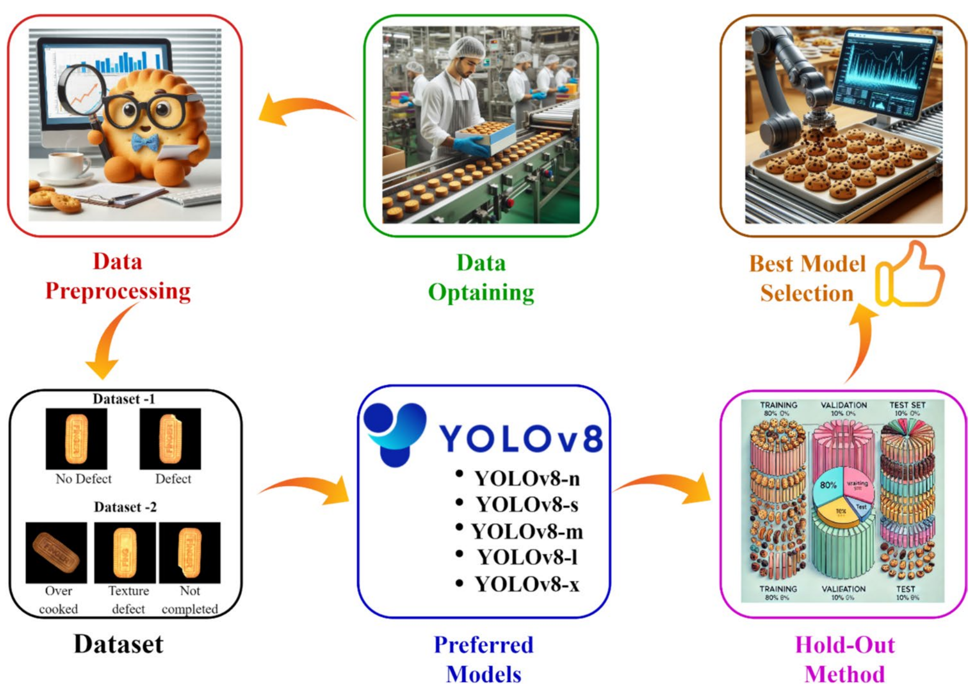


Fig. 1 Flow diagram of the project

Biscuit Dataset

In this study, images of finger biscuits, an industrial food product, were analyzed. The images were captured using an imaging system configured as illustrated in Fig. 2. A fixed lighting and camera setup was employed to minimize variations in factors such as angle, resolution, lighting, and background shadows. The biscuits were placed on a conveyor belt, and their images were captured in real time as they moved along the belt. The conveyor operated at a speed of approximately 0.5 m per second, and a high-speed camera, synchronized with the conveyor system, captured the images. The system automatically detected each sample and took a snapshot as it passed through the inspection area. Under experimental conditions, the setup processed approximately 1200 images per minute. The captured images were then transferred to a computer system via the integrated camera interface. This approach not only reduced processing time but also provided a cost-effective solution for image analysis.

The dataset has 4990 images of the same type of biscuits. The backgrounds are changed to black. The images are in RGB color, 2000 × 2000 pixels in size, and in "jpg" format. A representative sample of images by category is presented in Fig. 3. The total number of images in the dataset for each category is shown in Table 1. The biscuit dataset-1 consists of a total of two classes containing 4990 images, defect and no defect. Defect biscuits, on the other hand, are composed of three classes such that the biscuit dataset-2 is overcooked, not complete, and texture defect is included in Table 2. These numbers are included.

Feature Extraction

In the YOLO (You Only Look Once) model, properties are extracted by processing each image by a CNN. This CNN has been pre-trained on a large image dataset and has learned features that can be used for object detection (Khalaf & Abdulateef 2024). The "batch size" used during the training of the YOLO model refers to the number of samples used to update the parameters of the model at each optimization step. Batch size plays a decisive role on the model's performance and training time. A large batch size requires more memory usage, and this can lead to memory overuse and error generation, especially when GPU memory is limited (Hussain 2023). In terms of the training process, the big batch size usually reduces the training time because there are less frequent weight updates. However, this situation requires more computing power. In the context of generalization ability, small batch size contributes to the model's ability to generalize better by reducing the risk of over-learning, because each update contains more variations. In contrast, a large batch provides you with more stable and accurate updates, but it can increase the risk of over-learning. So, the best batch size can change based on things like the size of the dataset, how complex the model is, and the hardware capacity. Liu et al. (2023) mentioned that batch size values, which are usually determined as powers of 2, such as 8, 16,

Fig. 2 Data acquisition process

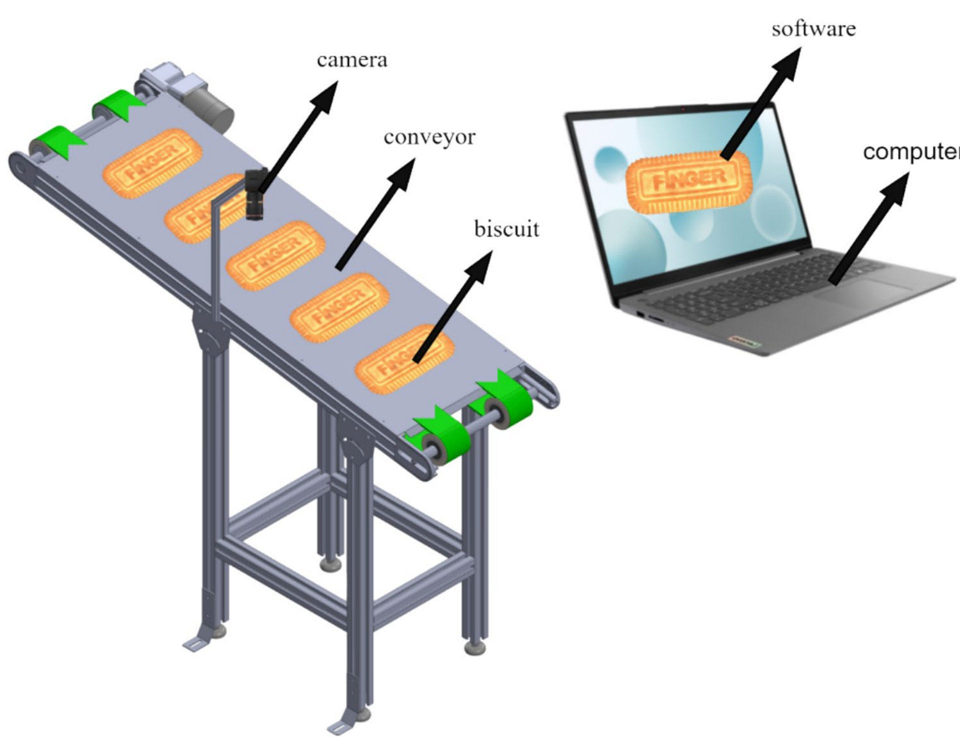


Fig. 3 Sample of the several categories of biscuit dataset. **a** No defect. **b** Overcooked. **c** Texture defect. **d** Not complete

32, and 64, are processed more efficiently by modern GPUs (Li et al. 2018). It is recommended to choose a batch such as 16 as the starting point for YOLO model training.

The YOLOv8 model used in the training process has been applied to you in batch 16, and there are train batch images in Fig. 4.

You Only Look Once

YOLO is a model developed by Joseph Redmon and Ali Farhadi in 2015 that uses CNNs for object detection and image segmentation. YOLOv2 has made significant contributions to the field by introducing bulk normalization and

Table 1 Distribution of the biscuit dataset-1

Class	Number of images
No defect	1648
Defect	3342
Total	**4990**

Table 2 Distribution of the biscuit dataset-2

Class	Number of images
Not complete	846
Overcooked	1042
Texture defect	1454
Total	**3342**

anchor boxes in 2016. YOLOv3, on the other hand, offered better performance in 2018 by coming with an improved mainnet and spatial pyramid pool. YOLOv4 has continued the developments in the field by introducing the Mosaic data augmentation method and a redesigned detection header in 2020. Following Meituan's adoption of YOLOV6 for autonomous delivery robots, YOLOv5 has been noted for its hyperparameter optimization and export flexibility. YOLOv7 has done studies on exposure estimation. YOLOv8, developed by Ultralytics, is an extremely up-to-date model that performs at a high level in many different areas such as detection, segmentation, exposure prediction, tracking, and classification, which shows its suitability for a wide range of image artificial intelligence applications (Gunawan et al. 2023). Yolov8 is a popular model in the fields of object detection, classification, and segmentation. It trains the entire image simultaneously and directly optimizes the detection performance. The initial convolution layer extracts features from the image or uses a fully connected layer for predicted probabilities (Terven et al. 2023). YOLOv8 was released in January 2023 by Ultralytics, the same company that made YOLOv5. The network architecture of the YOLOv8 program, whose model architecture is given in Fig. 5, consists of three main components: the spine, neck, and head. YOLOv8 has a new architecture, improved convolutional layers, the spine, and a more advanced perception (Cao et al. 2023). The YOLOv8 algorithm works by dividing an image into a grid of smaller regions and estimating a bounding box and class probabilities for each object (Terven & Cordova-Esparza 2023).

It comes in five versions: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra large). YOLOv8 can handle tasks like object detection, segmentation, exposure estimation, tracking, and classification. The characteristics of the YOLOv8 models are given in Table 3.

Hold-Out Method

The Hold-Out Method is a verification technique that is widely used in the training and evaluation process of

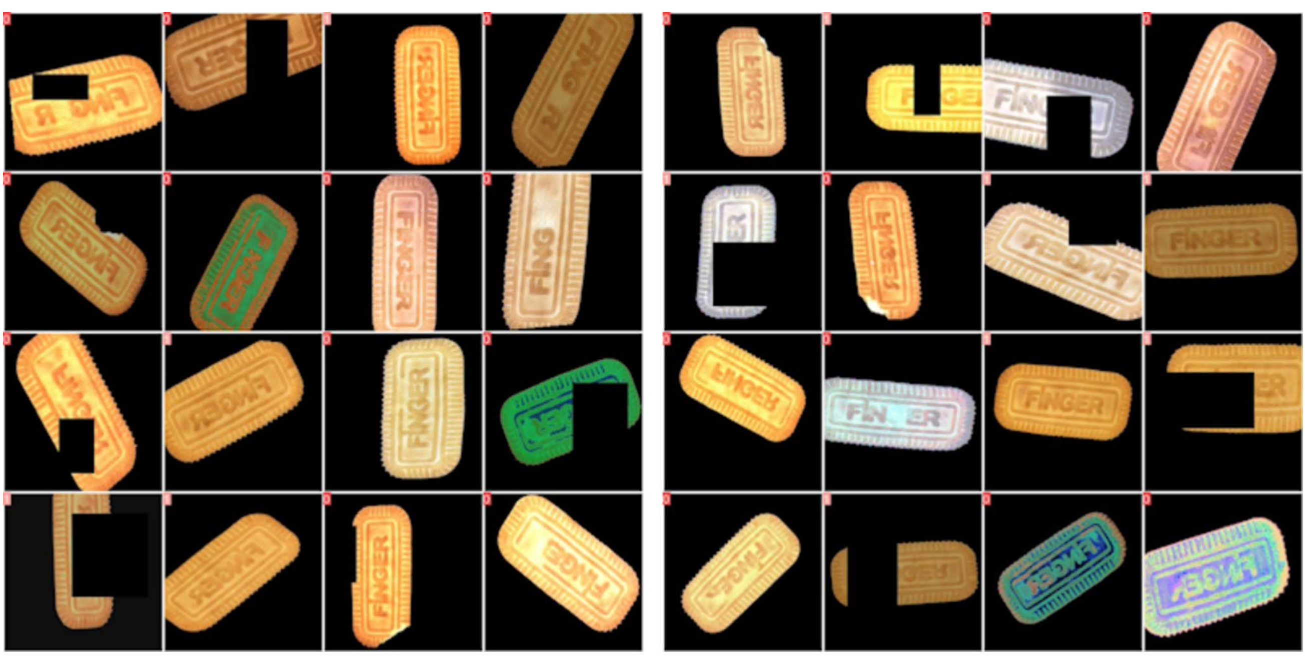


Fig. 4 Train batch

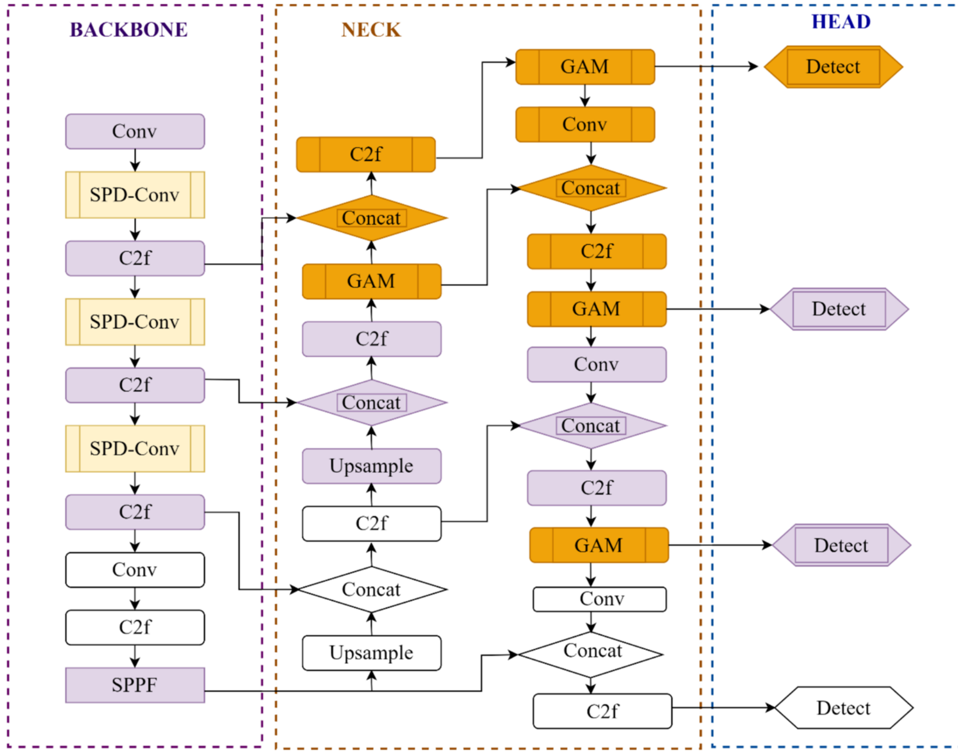


Fig. 5 YOLOv8 architecture

Table 3 YOLOv8 model variant

Model	Features	Use cases
YOLOv8n	Smallest and lightest model	Edge devices and real-time applications (Liu et al. 2023)
YOLOv8s	Small and compact	Mobile devices and embedded systems (Ma and Pang 2023)
YOLOv8m	Medium-sized model	General-purpose applications (Liu et al. 2023)
YOLOv8l	Large model	Server and powerful computer systems (Jiang et al. 2024)
YOLOv8x	Largest model	Comprehensive object detection projects and detailed data analysis (Jiang et al. 2024)

machine learning models. In this method, the dataset is randomly divided into two or three subsets: training dataset, validation dataset, and test dataset. These subsets are used to accurately evaluate the generalization ability and performance of the model (Pal & Patel 2020).

The dataset used in the study is a dataset containing two and three different classes. In order to evaluate the performance of the model, the dataset is divided into three separate sections: training 80%, verification 10%, and testing 10%. This division of the dataset is important for an objective evaluation of the generalization ability and accuracy of the model. The distribution of the dataset-1 and dataset-2 divided into training, verification, and test sets is shown in Fig. 6.

Confusion Matrix

A confusion matrix compares a dataset's predicted and actual class labels to assess the performance of a classification model (Koklu et al. 2021). The two-class confusion matrix is shown in Fig. 7 and the three confusion matrix representation is shown in Table 4 (Koklu & Ozkan 2020). TP is the number of samples that the model correctly classifies as positive (Ropelewska & Noutfia 2024). A model correctly classified TN as negative; an incorrectly classified FP as positive; and an incorrectly classified FN as negative is a model's incorrect classification (Çetin et al. 2024).

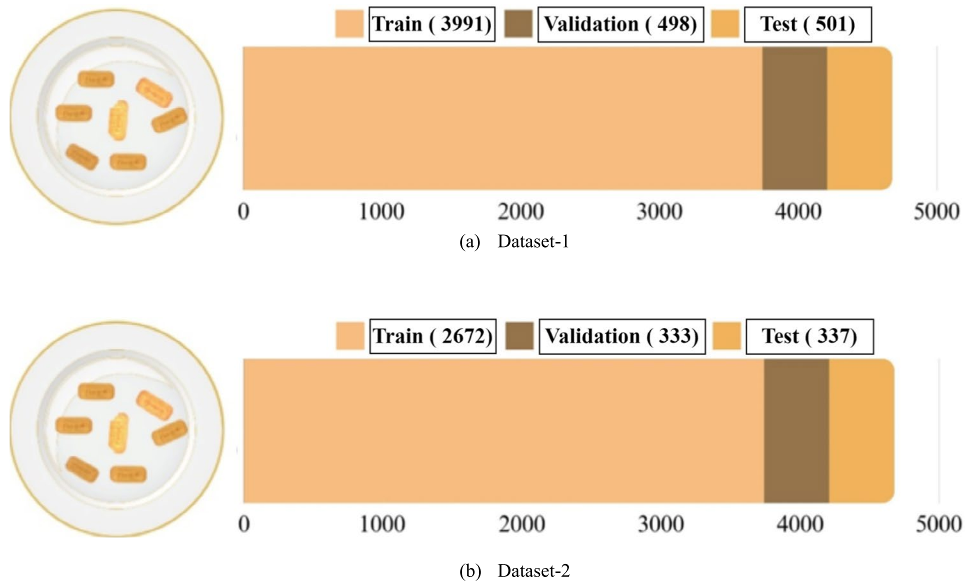


Fig. 6 Dataset-1–2 after hold-out (**a**, **b**). **a** Dataset-1, (**b**) dataset-2

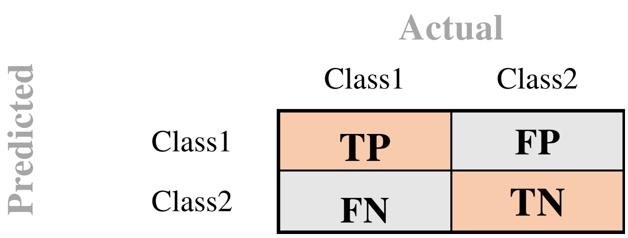


Fig. 7 Two class confusion matrix

Performance Metrics

Performance metrics are statistical measurements used to measure the effectiveness of a classification model (Gencturk et al. 2024). Performance metrics are evaluated through statistical criteria such as accuracy, train loss, and validation loss. Accuracy specifies the overlap rate of the class predicted by the model with the highest probability with the actual class (He et al. 2016). Accuracy's equation is given in Eq. 1 (Ropelewska et al. 2023).

$$Accuracy = \frac{Thenumberofcorrectprediction}{Totalnumberofestimate} \tag{1}$$

Train loss calculates the difference between the results predicted by the model and the actual results and is used to evaluate how decently the model has learned (Goodfellow et al. 2016).

Validation loss evaluates how the model performs on data other than the training dataset and helps determine the generalization ability of the model, is calculated at certain intervals during training, and is used to check whether the model is overfitting (Goodfellow et al. 2016).

Experımental Results

In this section, the results of the biscuit types classification experiments performed using YOLOv8 models are presented. The classification process was performed on the biscuit dataset using the YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x models, respectively. The accuracy and confusion matrix results obtained for each model are given in Tables 5 and 6, Tables 6 and 7, and the graphical results are given in Table 8.

Table 4 Confusion matrix and representation for multiple classes

		Actual				
		C_1	C_2	C_3	…	C_n
Predict	C_1	T_1	F_{12}	F_{13}	…	F_{1n}
	C_2	F_{21}	T_2	F_{23}	…	F_{2n}
	C_3	F_{31}	F_{32}	T_3	…	F_{3n}
	…	…	…	…	…	…
	C_n	F_{n1}	F_{n2}	F_{n3}	…	T_n

Table 5 Accuracy result of biscuit dataset-1

Algorithm	Epochs	Batch size	Accuracy (%)
YOLOv8-n	10	16	95.58
YOLOv8-s	10	16	95.98
YOLOv8-m	10	16	96.78
YOLOv8-l	10	16	96.78
YOLOv8-x	10	16	96.78

Table 6 Accuracy result of biscuit dataset-2

Algorithm	Epochs	Batch size	Accuracy (%)
YOLOv8-n	10	16	96.09
YOLOv8-s	10	16	96.69
YOLOv8-m	10	16	96.99
YOLOv8-l	10	16	96.39
YOLOv8-x	10	16	96.09

When Table 4 is examined, the accuracy results of the YOLO models obtained for the 2-class dataset of the defect and no defect types are seen. The YOLOv8-n model achieved the lowest classification accuracy with 95.58%. The YOLOv8-m-l-x models were the most successful model with 96.78% in 10 epochs..

When Table 6 was examined, the accuracy rate of the YOLOv8-m model for the YOLO models obtained for the 3-class dataset of the not complete, overcooked, and texture defect type achieved the highest classification accuracy with 96.99%. The YOLOv8-n model was the lowest model with 96.09%. YOLOv8-s, YOLOv8-l, and YOLOv8-x models obtained 96.69%, 96.39%, and 96.09% results respectively.

When examining the YoloV8-m model in Table 7, the model correctly predicted 323 true positives (TP) and 161 true negatives (TN). It made 3 false positive (FP) and 11 false negative (FN) predictions.

When examining the YoloV8-m model in Table 8, for the not completed class, it correctly predicted 82 instances, with only 1 instance misclassified as overcooked and 2 as texture defect. In the overcooked class, 101 instances were correctly predicted, with 1 instance misclassified as not completed and 3 as texture defect. Finally, for the texture defect class, the model accurately predicted 140 instances, with 1 instance misclassified as not completed and 2 as overcooked.

When the graphs in Fig. 8 are examined, the Yolov8-m model train/loss graph has decreased from 0.46 to 0.14. The validation/loss chart has decreased from 0.39 to 0.35, accuray 10. It was 96.78% for epoch.

When the graphs in Fig. 9 are examined, the train/loss graph of the Yolov8-m model has decreased from 0.65 to 0.10. The validation/loss chart has fallen from 0.63 to 0.58, accuracy 10. It was 96.99% for epoch.

Discussion

The results obtained with YOLOv8 models show high levels of accuracy in the classification of biscuit images. When the binary classification (defect and no defect) performance

Table 7 Confusion matrix results for biscuit dataset-1 test set

Yolov8_n		Actual: Defect	Actual: No Defect
Predicted	Defect	318	6
Predicted	No Defect	16	158

Yolov8_s		Actual: Defect	Actual: No Defect
Predicted	Defect	320	7
Predicted	No Defect	14	157

Yolov8_m		Actual: Defect	Actual: No Defect
Predicted	Defect	323	3
Predicted	No Defect	11	161

Yolov8_I		Actual: Defect	Actual: No Defect
Predicted	Defect	324	7
Predicted	No Defect	10	157

Yolov8_x		Actual: Defect	Actual: No Defect
Predicted	Defect	321	2
Predicted	No Defect	13	162

Table 8 Confusion matrix results for biscuit dataset-2 test set

Yolov8_n		Actual: Not Completed	Actual: Over Cooked	Actual: Texture Defect
Predicted	Not Completed	81	1	1
	Over Cooked	1	101	6
	Texture Defect	2	2	138

Yolov8_s		Actual: Not Completed	Actual: Over Cooked	Actual: Texture Defect
Predicted	Not Completed	81	0	0
	Over Cooked	1	101	5
	Texture Defect	2	3	140

Yolov8_m		Actual: Not Completed	Actual: Over Cooked	Actual: Texture Defect
Predicted	Not Completed	82	1	2
	Over Cooked	1	101	3
	Texture Defect	1	2	140

Yolov8_l		Actual: Not Completed	Actual: Over Cooked	Actual: Texture Defect
Predicted	Not Completed	81	0	0
	Over Cooked	1	102	4
	Texture Defect	2	2	141

Yolov8_x		Actual: Not Completed	Actual: Over Cooked	Actual: Texture Defect
Predicted	Not Completed	81	1	2
	Over Cooked	1	101	4
	Texture Defect	2	2	139

was evaluated, the YOLOv8-n model achieved the lowest classification accuracy with 95.58%. YOLOv8-m-l-x models achieved the highest accuracy of 96.78% with 10 epochs. The other model, YOLOv8-s achieved accuracy rates of 95.98% respectively. These results show that the models showed similar performances in distinguishing between defect and no defect biscuits, but there were small differences.

In three category classifications (not complete, overcooked, texture defect), the YOLOv8-m model has exhibited the best performance with an accuracy rate of 96.99%. The YOLOv8-n model had the lowest accuracy with 96.09%. The YOLOv8-s, YOLOv8-l, and YOLOv8-x models achieved accuracy rates of 96.69%, 96.39%, and 96.09% respectively. This shows that all models are effective, but YOLOv8-m has a slight superiority in more detailed error categories.

The confusion matrix analysis in Table 6 shows that YOLO models make accurate predictions for the most part. In binary classification, YOLOv8-n model performed the most misclassifications with 22 errors and especially confused no defect biscuits as defects. In the three category classifications, the most confused category was Texture defect, which highlights the difficulty in distinguishing this specific type of error.

The training loss, verification loss, accuracy, and graphs of the models are examined in Fig. 8. Examining the training

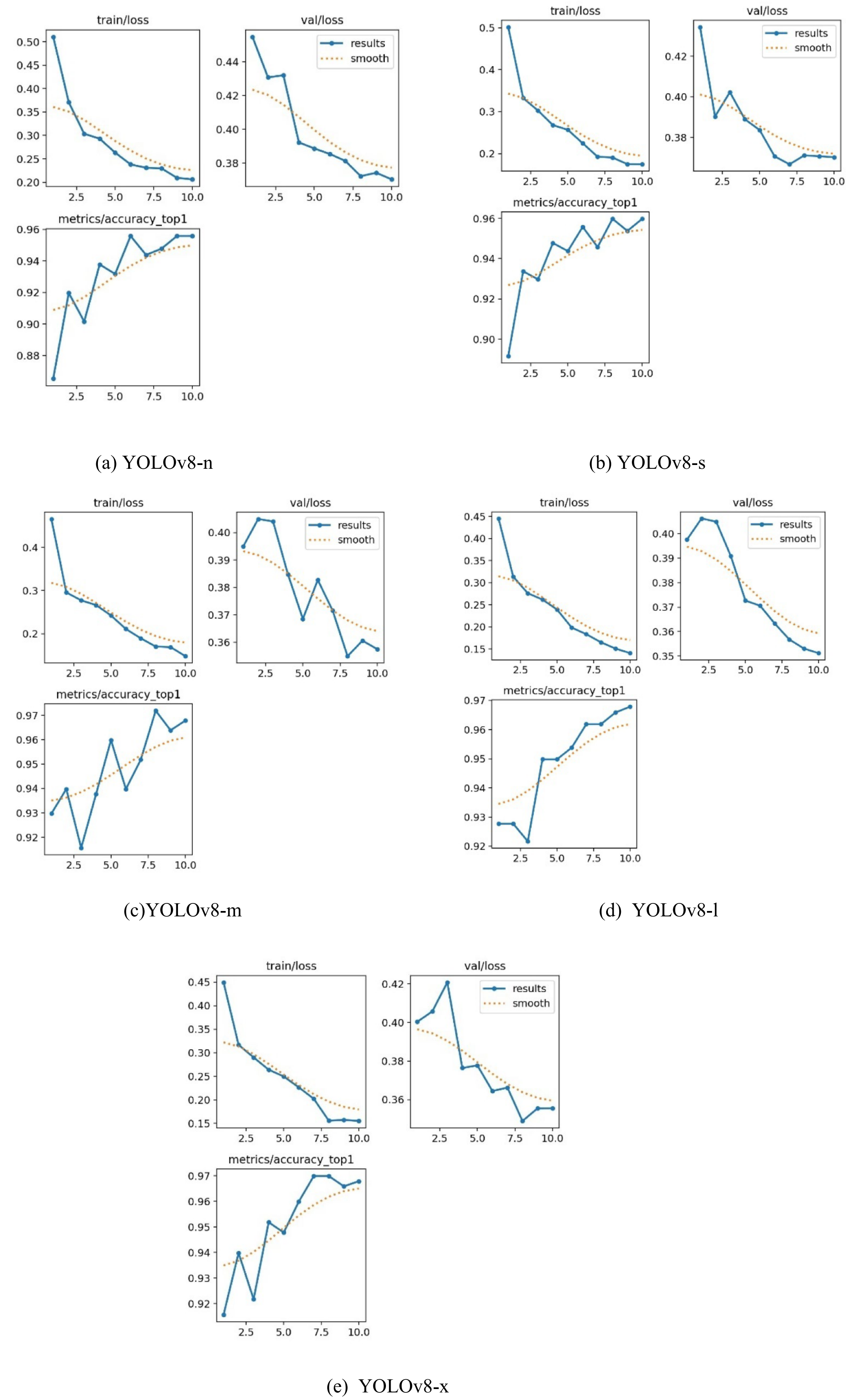
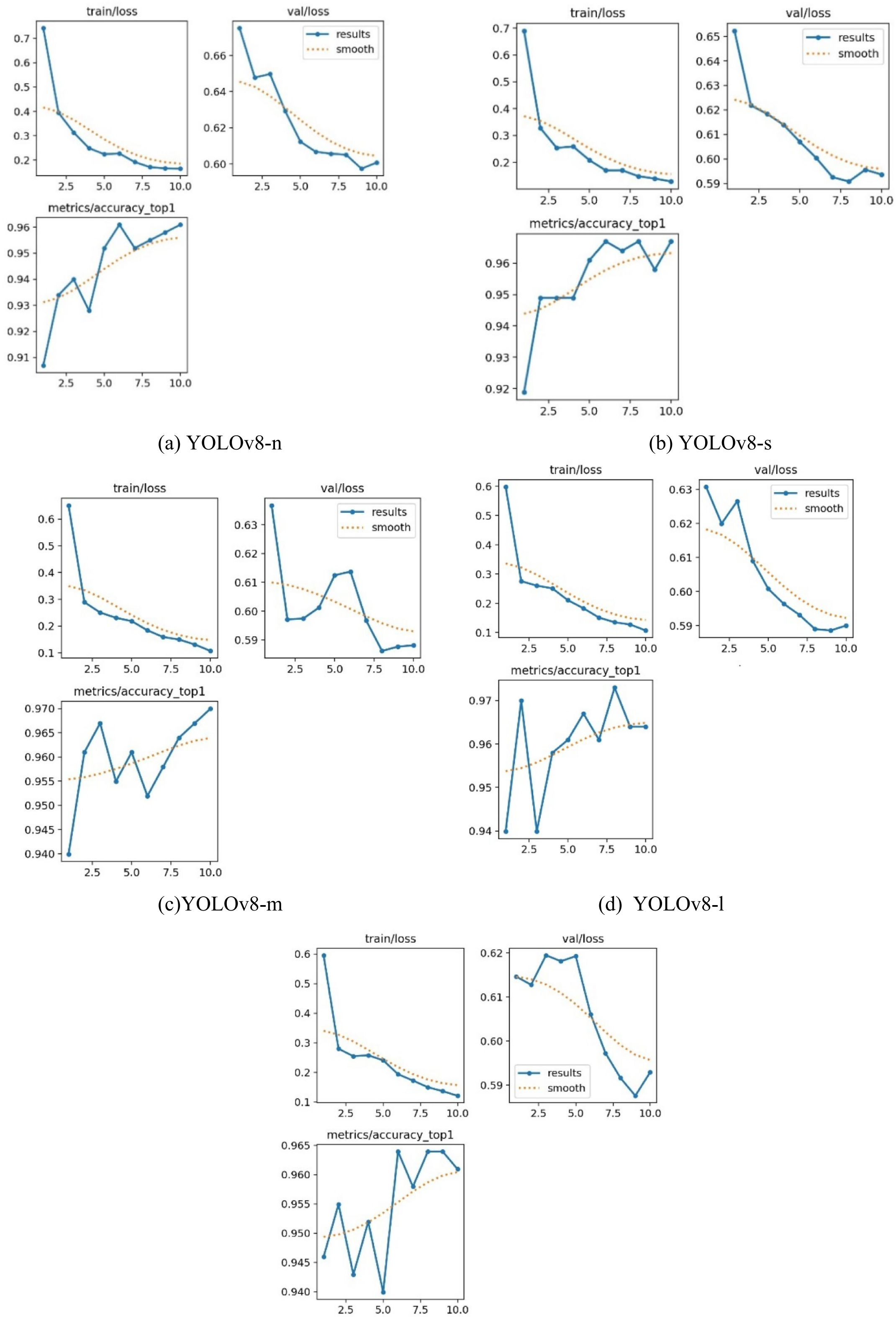


Fig. 8 Results of YOLOv8 models for biscuit dataset-1. **a** YOLOv8-n, (**b**) YOLOv8-s, (**c**) YOLOv8-m, (**d**) YOLOv8-l, (**e**) YOLOv8-x



(a) YOLOv8-n

(b) YOLOv8-s

(c)YOLOv8-m

(d) YOLOv8-l

(e) YOLOv8-x

Fig. 9 Results of YOLOv8 models for biscuit dataset-2. **a** YOLOv8-n, (**b**) YOLOv8-s, (**c**) YOLOv8-m, (**d**) YOLOv8-l, (**e**) YOLOv8-x loss, validation loss, accuracy, and graphs of the models in Fig. 8, it is seen that the loss decreases as the epoch number increases for the most successful YOLOv8-m model. While the training loss was ~ 0.6 at the beginning, it decreased continuously to almost 0.1 for/along 10 epochs. It is seen that it has decreased to almost 0.1 level in epoch. This decrease shows that the model is getting better on the training data and reducing its errors. In the same way, the loss of verification decreases as the number of epochs increases. The verification loss, which was 0.6 at the beginning, it decreased continuously to almost 0.1 for/along 10 epochs. It is seen that it has fallen to 0.5 level in epoch.

In Fig. 9, the not complete, overcooked, texture defect dataset decreased regularly from 0.65 to 0.1 depending on the epoch number of training loss when examining the graphs of the models and looking at the YOLOv8-m model results. This shows that the model is getting better on the training data and reducing its errors. The loss of verification also decreases as the number of epochs increases. The verification loss, which was 0.63 at the beginning, is 10. It is seen that it has fallen to 0.58 level in epoch. It is understood that the verification loss is due to the fact that the model receives such a small decrease in 2 different ways when labeling the same biscuit, which is not complete and texture defect in the verification part of the model. Figure 10 also contains this situation.

The fact that the biscuits are robust, hygienic, and perfect is of great importance in terms of food safety and consumer satisfaction. Robust biscuits protect the physical integrity of the product by preventing damage that may occur during production and transportation processes, which meets the expectations of the consumer. Hygienic production conditions prevent foodborne diseases by minimizing the risk of microbial contamination and protect consumer health. Perfect products strengthen the brand image with their aesthetic and satisfactory features and increase consumer loyalty. In addition, the maintenance of high-quality standards ensures compliance with national and international food safety regulations, which helps manufacturers to maintain and expand their Sunday share. Therefore, robustness, hygiene, and perfection in biscuit production are indispensable criteria for both public health and commercial success. The detection of defective industrial products on production lines has been carried out using various sensor technologies and image processing methods. Image processing-based systems, utilizing high-resolution cameras and artificial intelligence algorithms, have been implemented to accurately identify defective products. Deep learning models, particularly those supported by algorithms such as CNN, are capable of detecting surface defects with high accuracy, thereby improving the effectiveness of quality control processes. Sensor-based systems also measure the physical properties of products to separate defective items.

The removal of detected defective products from the production line has been achieved through laser systems, robotic arms, and ejection systems. Laser systems identify defects on the product surface and initiate the appropriate sorting action. Robotic arms efficiently and accurately remove defective products from the production line. Ejection systems, which push defective items off the conveyor, are

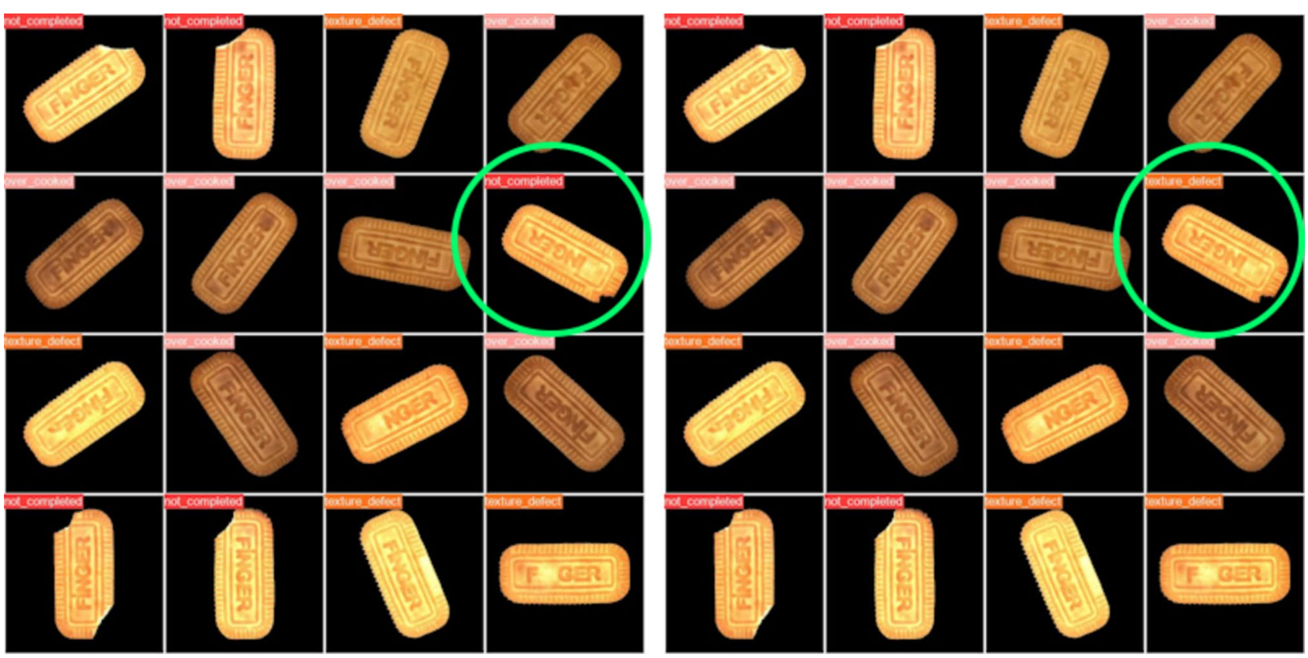


Fig. 10 Validation loss labels

used as mechanisms to separate faulty products. These technologies play key roles in automated production processes, enhancing production speed while reducing labor costs.

However, there is a lack of studies in the literature regarding the rate of defective products on industrial production lines. Discussions with manufacturing companies revealed that there are no existing systems in place to monitor this rate. While most manufacturers focus on quality control processes, systems to determine how frequently defective products occur have not been developed. This situation highlights the importance of data collection and analysis processes to make quality control in industrial production more efficient.

Conclusıon

The prepared study successfully demonstrates the application of YOLOv8 models in the automatic classification of biscuits in the food industry and highlights the potential for improving quality control processes. The models have achieved high accuracy rates and the YOLOv8-l and YOLOv8-m models have shown the best performance in different classification tasks. The small performance differences between the models suggest that selection can be made according to specific classification needs and computational resources.

The classification approach adopted in this study and the strong performance of YOLOv8 models can significantly contribute to ensuring higher quality standards in biscuit production. By expanding the potential and application areas of YOLOv8 models in the food industry, it has taken an important step towards automating quality control in production processes. Such future studies can promote the production of higher quality and reliable food products and improve the overall efficiency of the food industry.

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Data Availability https://www.muratkoklu.com/datasets/Finger_Biscuit_Datasets.zip (accessed on December 18, 2024).

Declarations

Conflict of Interest The authors declare no competing interests.

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